UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

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QUESTION 1

a) Solve the inequality $3x^2 + 24x + 45 > 0$.

(5 marks)

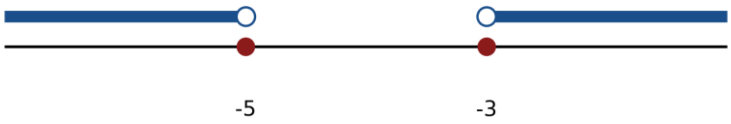


Figure: critical values.

b) Solve the inequality $|x + 1| <= 7$. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Solve the equation $|-5x - 2| = 3$.

(4 marks)

QUESTION 2

a) Apply graph-transformation techniques to sketch $y = -3(x - 1)^2 + 5$. State the vertex and the y-intercept.

(5 marks)

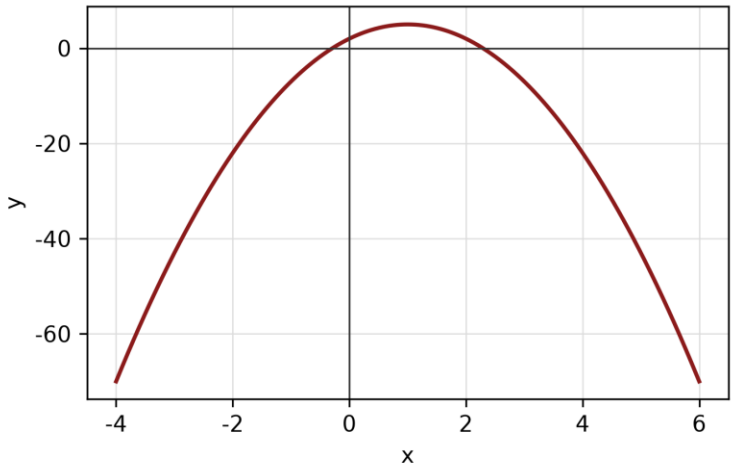


Figure: transformed graph.

b) Determine algebraically whether $f(x) = x^4 + x^2 - 5$ is even, odd or neither. State the symmetry of its graph.

(4 marks)

c) Find the standard equation of the parabola with focus (-1, -1) and directrix y = -5. State its vertex.

(5 marks)

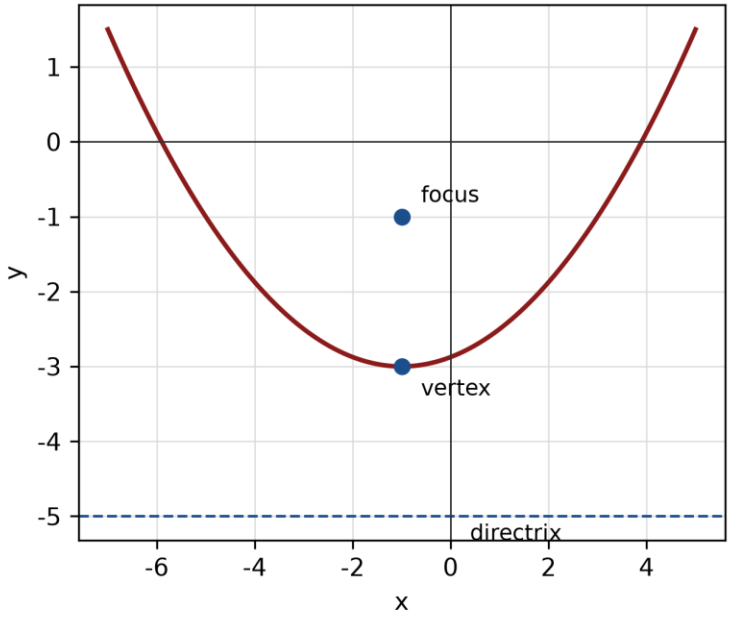


Figure: parabola with focus and directrix.

QUESTION 3

a) Given $z_1 = -2 - 4i$ and $z_2 = -5 + 6i$, find $z_1 \times z_2$ in the form a + bi.

(4 marks)

b) Evaluate i^37 and simplify completely.

(3 marks)

c) Show that x + 3 is a factor of f(x) = -x^3 - 4x² - 6x - 9 by using the factor theorem. Hence factorise f(x) completely.

(5 marks)

QUESTION 4

a) Solve the simultaneous equations
x + 4y = -13
-x + y = -2.

(4 marks)

b) Write a system of inequalities that describes the shaded triangular region with vertices (0, 0), (3, 0) and (0, 3).

(5 marks)

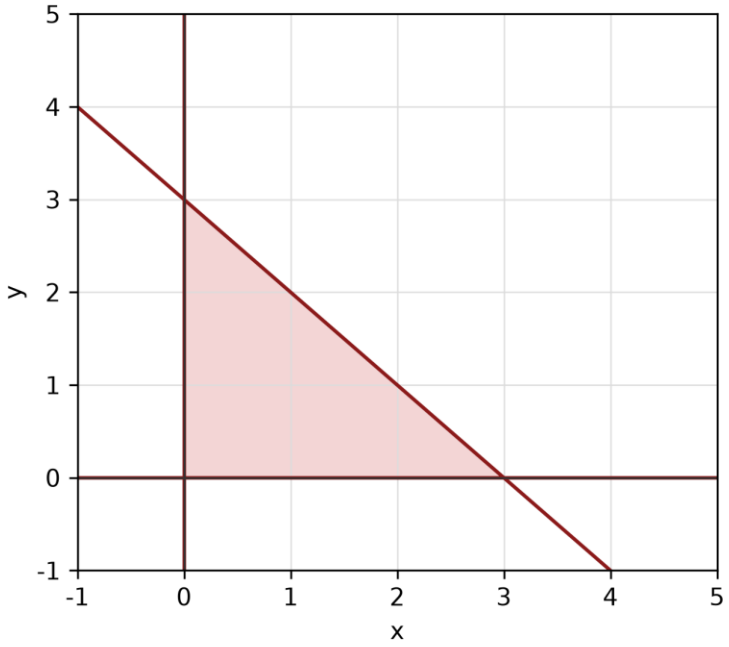


Figure: shaded triangular region.

c) Write $x^2 - 6x + y^2 - 10y + 30 = 0$ in standard form. Hence state the centre and radius and sketch the circle.

(5 marks)

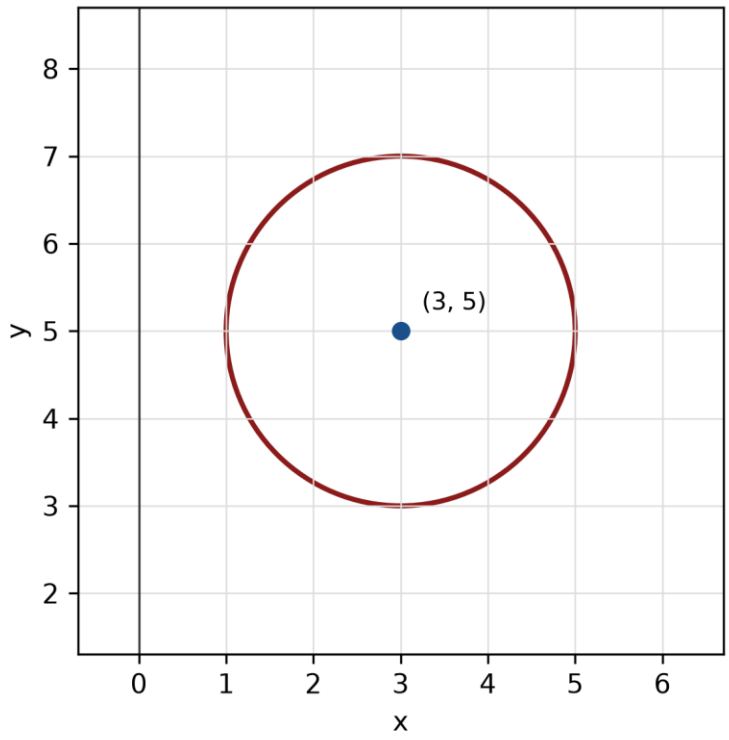


Figure: circle.

QUESTION 5

a) For y = -2 sin(3x + 60) + 1, state the amplitude, period and phase shift, then sketch the

graph for -180° ≤ x ≤ 180°.

(5 marks)

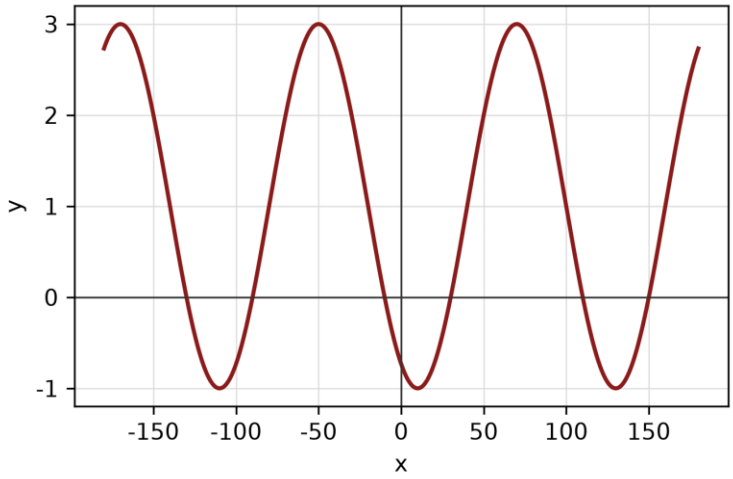


Figure: trigonometric graph.

b) Convert 135° to radians, and convert 2π/3 radians to degrees.

(3 marks)

c) Find the area of triangle ABC with a = 11 cm, b = 10 cm and included angle C = 45°.

(4 marks)

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$
$\sin(A + B) = \sin A \cos B + \cos A \sin B$
$\cos(A + B) = \cos A \cos B - \sin A \sin B$
$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$
$\cos 2A = \cos^2 A - \sin^2 A$
$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
$\sin 2A = 2 \sin A \cos A$
$\tan^2 A + 1 = \sec^2 A$
$\tan 2A = 2 \tan A / (1 - \tan^2 A)$
Arc length, $s = r\theta$
Area of sector = $\frac{1}{2}r^2\theta$
$a / \sin A = b / \sin B = c / \sin C$
$a^2 = b^2 + c^2 - 2bc \cos A$
Area of triangle = $\frac{1}{2}ab \sin C$

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